MAC21 — The Complete 4-Mark Bank

Every 4-mark question asked, worked out. 45 questions with full solutions. You face five 4-markers and choose the easier of two in each unit — that is **20 marks**, the cheapest big block in the paper after the 2-markers.

Unlike the 2-markers, these are short calculations — you must show working, not just the answer. Every number below was computed and verified. **n×** = times asked across the 6 papers; **LIKELY** = not yet asked but on the portion.
Budget 6 minutes each. If one runs past 8, write the setup and move on.

UNIT I — Q1(b) or Q2(b)

#	QUESTION	WORKED ANSWER	SEEN
1	Show $f(x,y) = xy(1-x-y)$ is **maximum at (1/3, 1/3)**	$f_x = y(1-2x-y) = 0$, $f_y = x(1-x-2y) = 0$ → stationary points $(0,0)$, $(1,0)$, $(0,1)$, $(1/3,1/3)$ $r = -2y$, $s = 1-2x-2y$, $t = -2x$ At $(1/3,1/3)$: $r = -2/3$, $s = -1/3$, $t = -2/3$, so $rt - s^2 = 4/9 - 1/9 = 1/3 > 0$ and $r < 0$ **∴ maximum, and $f_{max} = 1/27$**	3×
2	Examine $x^3 + y^3 - 3axy$ for extreme values	$f_x = 3x^2-3ay = 0$, $f_y = 3y^2-3ax = 0$ → $(0,0)$ and (a,a) $r = 6x$, $s = -3a$, $t = 6y$ → $rt - s^2 = 36xy - 9a^2$ At $(0,0)$: $-9a^2 < 0$ → **saddle point** At (a,a): $27a^2 > 0$, $r = 6a$ → **minimum if a>0, maximum if a<0; $f = -a^3$**	1×
3	Expand a_x in powers of x, first three terms	$a_x = e_x$ Ộṭụ ạ ... use $f(x)=a_x$, $f'=a_x \log a$, $f''=a_x(\log a)^2$; at 0: $1, \log a, (\log a)^2$ **$a_x = 1 + x \log a + x^2(\log a)^2/2! + \ldots$**	1×
4	Expand $\sin^{-1}x$ in powers of x	$d/dx(\sin^{-1}x) = (1-x^2)^{-\frac{1}{2}} = 1 + x^2/2 + 3x^4/8 + \ldots$ (binomial) Integrate term by term: **$\sin^{-1}x = x + x^3/6 + 3x^5/40 + \ldots$** *Odd function — every even power is zero. Say so.*	1×
5	Expand $\tan^{-1}x$ in powers of $(x-1)$ to 4th degree	Taylor about $a=1$. $f(1)=\pi/4$; $f'=1/(1+x^2)$ → 1/2; $f''=-2x/(1+x^2)^2$ → −1/2; f''' → 1/2; f^{iv} → 0 **$\tan^{-1}x = \pi/4 + (x-1)/2 - (x-1)^2/4 + (x-1)^3/12 + 0\cdot(x-1)^4$** *The 4th-degree term genuinely vanishes.*	1×
6	**N-R system, ONE iteration**: $x^2+y^2=x$, $x^2-y^2=y$ from $(0.8, 0.4)$	$f = 0.0000$, $g = 0.0800$; $J = [[0.6, 0.8],[1.6, -1.8]]$ Solve $J[h,k]^T = -[f,g]^T$ → $h = -0.027119$, $k = 0.020339$ **$x_1 = 0.772881$, $y_1 = 0.420339$**	1×
7	**N-R system, ONE iteration**: $x^2+y=11$, $y^2+x=7$ from $(3.5, -1.8)$	$f = -0.5500$, $g = -0.2600$; $J = [[7, 1],[1, -3.6]]$ $h = 0.085496$, $k = -0.048473$ **$x_1 = 3.585496$, $y_1 = -1.848473$**	1×
8	Find a **negative root** of $x\,e_x - \sin x = 0$ to three decimals	Bracket first: $f(-2.9)=+0.080$, $f(-3.0)=-0.008$ → root in $(-3, -2.9)$ $f' = e_x(1+x) - \cos x$; from $x_0=-3$: $x_1=-2.990745$, $x_2=-2.990735$ **root = −2.991** *Starting near 0 collapses to the trivial root x=0.*	2×
9	Can N-R solve $x\,e^-{}_x = 0$ near $x=2$? Justify	$f'(x) = e^-{}_x(1-x)$, which is **0 at x=1** and $-e^{-2} \approx -0.135$ at $x=2$ while $f(2)=0.271$. The correction f/f' is large and pushes x away from the root at $x=0$. **N-R diverges here — it fails when f′ is zero or small near the start.**	1×
10	Expand $e_x \log x$ / $\sqrt{(1+\sin 2x)}$ — other expansions	Same Taylor/Maclaurin machinery: compute f, f', f'', f''' at the point, substitute. For $\sqrt{(1+\sin 2x)}$: note $1+\sin 2x = (\sin x + \cos x)^2$, so it is just $\sin x + \cos x$ → **$1 + x - x^2/2 - x^3/6 + x^4/24$**	LIKELY

UNIT II — Q3(b) or Q4(b)

#	QUESTION	WORKED ANSWER	SEEN
11	**Bungee jumper**, $m=68.1$, $c=0.25$, $g=9.8$, Euler, 1-second steps	$dv/dt = 9.8 - 0.003671v^2$ ($c/m = 0.25/68.1$) $t=0$: $v=0$, $f=9.8000$ → $v(1)=9.8000$ $t=1$: $v=9.8$, $f=9.4474$ → $v(2)=19.2474$ $t=2$: $v=19.2474$, $f=8.4400$ → **$v(3) = 27.6874$ m/s**	2×
12	**Euler's method**: $dy/dx - 2y = 3e_x$, $y(0)=0$, at $x=0.2$, $h=0.1$	$f(x,y) = 2y + 3e_x$ $y(0.1) = 0 + 0.1(0 + 3) = 0.300000$ $y(0.2) = 0.3 + 0.1(0.6 + 3e^{0.1}) =$ **0.691551**	2×
13	Same DE with $y(1)=0$, at $x=1.1$, $h=0.05$	$y(1.05) = 0 + 0.05(0 + 3e^1) = 0.407742$ $y(1.10) = 0.407742 + 0.05(0.815484 + 3e^{1.05}) =$ **0.877164**	1×

14	**Taylor's series method**: $dy/dt = e^t - 2y$, $y(0)=1$, at $t=0.1$, 3rd degree	$y' = e^t - 2y \to -1$; $y'' = e^t - 2y' \to 3$; $y''' = e^t - 2y'' \to -5$ $y(0.1) = 1 + 0.1(-1) + (0.01/2)(3) + (0.001/6)(-5)$ **$y(0.1) = 0.914167$**	2×
15	**Newton's law of cooling**: object 300°F, room 80°F, 250°F after 10 min — temperature after 20 min?	$T = 80 + 220e^{-kt}$. At $t=10$: $170 = 220e^{-10k} \to e^{-10k} = 17/22$ At $t=20$: $T = 80 + 220(e^{-10k})^2 = 80 + 220(17/22)^2$ **$T(20) = 211.36$ °F** *No need to find k — square the ratio.*	2×
16	**Orthogonal trajectory of cardioids** $r = a(1-\cos\theta)$, given $dr/d\theta = r\cot(\theta/2)$	Replace $dr/d\theta \to -r^2(d\theta/dr)$: $-r(d\theta/dr) = \cot(\theta/2)$ $\Rightarrow dr/r = -\tan(\theta/2)d\theta \to \log r = 2\log\cos(\theta/2) + \log c$ **$r = c\cos^2(\theta/2) = C(1 + \cos\theta)$** — the reflected cardioid	1×
17	If $r^n = k\cos(n\theta)$, find the **DE of that family**	Take logs: $n\log r = \log k + \log\cos(n\theta)$ Differentiate w.r.t. θ: $n(1/r)(dr/d\theta) = -n\tan(n\theta)$ **$(1/r)(dr/d\theta) = -\tan(n\theta)$, i.e. $dr/d\theta + r\tan(n\theta) = 0$**	1×
18	**Orthogonal trajectory of** $r = b\sin\theta$	Eliminate b: $(1/r)(dr/d\theta) = \cot\theta$. Swap → $(1/r)(dr/d\theta) = -\tan\theta$ $dr/r = -\tan\theta\, d\theta \to \log r = \log\cos\theta + \log c$ **$r = c\cos\theta$**	2×
19	**Orthogonal trajectories, Cartesian** — general 4-mark form	1. Differentiate, eliminate the parameter. 2. Replace $dy/dx \to -dx/dy$. 3. Separate and integrate. *e.g.* $x^2+y^2=2ax$ *→ DE* $2xy\,y' = y^2-x^2$ *→ swap →* **$x^2+y^2 = 2by$**	LIKELY
20	**Modified Euler**, one stage (4-mark version)	Predictor $y_1^{(0)} = y_0 + h\,f(x_0,y_0)$, then corrector $y_1^{(k+1)} = y_0 + (h/2)[f(x_0,y_0) + f(x_1, y_1^{(k)})]$ twice. *Keep* $f(x_0,y_0)$ *fixed; only the second term updates.*	LIKELY

UNIT III — Q5(b) or Q6(b)

#	QUESTION	WORKED ANSWER	SEEN
21	Prove $(1/(D+a))X = e^{-ax}\int X e^{ax}\,dx$ **the safest 4 marks in the paper**	Let $y = (1/(D+a))X$. Then $(D+a)y = X$, i.e. $dy/dx + ay = X$. Linear first-order; integrating factor e^{ax}: $e^{ax}(dy/dx) + a\,e^{ax}y = X\,e^{ax} \to d(y\,e^{ax})/dx = X\,e^{ax}$ Integrate: $y\,e^{ax} = \int X e^{ax}dx \to$ **$y = e^{-ax}\int X e^{ax}dx$** ■	4×
22	General solution if the A.E. is $\lambda^3(\lambda+4)^2(\lambda^2+2\lambda+5)^2 = 0$	Roots: 0 (three times), -4 (twice), $-1\pm2i$ (each twice) **$y = (C_1+C_2x+C_3x^2) + (C_4+C_5x)e^{-4x} + e^{-x}[(C_6+C_7x)\cos 2x + (C_8+C_9x)\sin 2x]$** *Repeated complex roots get the extra x factor too.*	1×
23	Show $y^{iv} - k^4y = 0$ has solution $C_1\cos kx + C_2\sin kx + C_3\cosh kx + C_4\sinh kx$	A.E.: $m^4 - k^4 = (m^2-k^2)(m^2+k^2) = 0 \to m = \pm k, \pm ik$ $y = Ae^{kx} + Be^{-kx} + C\cos kx + D\sin kx$ Since $e^{kx} = \cosh kx + \sinh kx$ and $e^{-kx} = \cosh kx - \sinh kx$, the first two combine into **$C_3\cosh kx + C_4\sinh kx$**. ■	1×
24	Solve the BVP $y'' + 4y' + 4y = 0$, $y(1)=0$, $y'(0)=-1$	A.E. $(m+2)^2=0 \to y = (A+Bx)e^{-2x}$; $y' = (B - 2A - 2Bx)e^{-2x}$ $y'(0) = B - 2A = -1$; $y(1) = (A+B)e^{-2} = 0 \Rightarrow B = -A$ $\to -3A = -1$, $A = 1/3$, $B = -1/3 \to$ **$y = (1 - x)e^{-2x}/3$**	1×
25	Solve the IVP $y'' - y = 0$, $y(0)=3$, $y'(0)=3$	$m^2-1=0 \to y = Ae^x + Be^{-x}$. $A+B=3$, $A-B=3 \to A=3, B=0$ **$y = 3e^x$**	1×
26	Find the P.I. of $y'' - y' = x^2 + 2x + 4$	$y_p = 1/(D^2-D)\cdot(x^2+2x+4) = -(1/D)\cdot(1-D)^{-1}(x^2+2x+4)$ $(1-D)^{-1} = 1 + D + D^2 + \dots$ → applied: $(x^2+2x+4) + (2x+2) + 2 = x^2+4x+8$ Then $-(1/D)$ means integrate and negate: **$y_p = -(x^3/3 + 2x^2 + 8x)$**	1×
27	Solve the Legendre equation $(2x+3)^2y'' + 2(2x+3)y' - 12y = 0$	Put $2x+3 = e^z$, $a=2$: $4D(D-1)y + 4Dy - 12y = 0 \to 4D^2y = 12y$ $m^2 = 3 \to m = \pm\sqrt{3} \to y = C_1e^{\sqrt{3}z} + C_2e^{-\sqrt{3}z}$ **$y = C_1(2x+3)^{\sqrt{3}} + C_2(2x+3)^{-\sqrt{3}}$** *Do not drop the* $a^2 = 4$.	1×
28	Write the C.F. given roots $-3\pm i$, $2, 2, 2$	**$y = e^{-3x}(C_1\cos x + C_2\sin x) + (C_3 + C_4x + C_5x^2)e^{2x}$**	1×
29	C.F. of a 4th-order LDE: two roots imaginary, two real distinct	Roots $\alpha\pm i\beta$, m_1, m_2 → **$y = e^{\alpha x}(C_1\cos\beta x + C_2\sin\beta x) + C_3e^{m_1x} + C_4e^{m_2x}$**	1×
30	P.I. for a standard case — general 4-mark form	$e^{ax} \to e^{ax}/f(a)$; if $f(a)=0$ multiply by x and differentiate the denominator. $\sin/\cos ax$ → put $D^2 = -a^2$. $e^{ax}V \to e^{ax}(1/f(D+a))V$.	LIKELY

UNIT IV — Q7(b) or Q8(b)

#	QUESTION	WORKED ANSWER	SEEN

31	**Missing terms**: $x = 45,50,55,60,65$; $y = 3.0, -, 2.0, -, -2.4$	3 known values → quadratic fits → **both** third differences vanish: $\Delta^3 y_0 = 3a + b - 9 = 0$, $\Delta^3 y_1 = -a - 3b + 3.6 = 0$ **$y(50) = 2.925$, $y(60) = 0.225$** *Check: 2nd differences all -0.85. Show this.*	2×
32	**Missing terms**: $x = 1 \ldots 7$; $y = 2, 5, -, 13, 22, -, 63$	5 known values → quartic fits → both **fifth** differences vanish. Equivalently, fit the quartic through $(1,2)(2,5)(4,13)(5,22)(7,63)$: $P(x) = x^4/60 + 2x^3/15 - 71x^2/60 + 161x/30 - 7/3$ **$y(3) = 121/15 = 8.067$, $y(6) = 113/3 = 37.667$**	1×
33	**Trapezoidal**: $\int_0^1 dx/(1+x^2)$, 4 equal strips	$h = 0.25$; $y = 1, 0.941176, 0.8, 0.64, 0.5$ $I = (0.25/2)[(1+0.5) + 2(0.941176+0.8+0.64)]$ **$I = 0.782794$** (exact $\pi/4 = 0.785398$; $4I \approx 3.1312$)	1×
34	**Trapezoidal**: $\int_0^1 e^x\, dx$, steps of 0.2	$h = 0.2$; $y = 1, 1.221403, 1.491825, 1.822119, 2.225541, 2.718282$ $I = (0.2/2)[(1+2.718282) + 2(1.221403+1.491825+1.822119+2.225541)]$ **$I = 1.724006$** (exact $e-1 = 1.718282$)	1×
35	**Backward difference table** for $y = 2x^2 - x + \cos 2$ on $(2,3)$, $h=0.2$; find $\nabla^2 y_3$	y: 5.583853, 7.063853, 8.703853, 10.503853, 12.463853, 14.583853 ∇: 1.48, 1.64, 1.80, 1.96, 2.12 ∇^2: 0.16, 0.16, 0.16, 0.16 **$\nabla^2 y_3 = 0.16$** *y is quadratic, so 2nd differences are constant $= 2ah^2 = 2(2)(0.04)$.*	1×
36	Estimate $\cosh(0.56)$ from $x = 0.5, 0.6, 0.7, 0.8$	Δ: 0.0579, 0.0697, 0.0822; Δ^2: 0.0118, 0.0125; Δ^3: 0.0007 $p = (0.56-0.5)/0.1 = 0.6$ → Newton forward: $1.1276 + 0.6(0.0579) + (0.6)(-0.4)/2 \cdot (0.0118) + \ldots$ **$\cosh(0.56) \approx 1.16096$** (true 1.16094)	1×
37	**Lagrange interpolation**: $f(1)$ from $(-1,9), (0,5), (2,3)$	$P(x) = 9 \cdot x(x-2)/3 + 5 \cdot (x+1)(x-2)/(-2) + 3 \cdot (x+1)x/6$ Simplify: $P(x) = x^2 - 3x + 5$ → **$f(1) = 3$**	1×
38	**Lagrange inverse**: find x when $y=15$ from $(6,13), (9,14), (11,16)$	Swap roles: $x = 6 \cdot (15-14)(15-16)/((13-14)(13-16)) + 9 \cdot (15-13)(15-16)/((14-13)(14-16)) + 11 \cdot (15-13)(15-14)/((16-13)(16-14))$ $= 6(-1/3) + 9(1) + 11(1/3)$ → **$x = 32/3 = 10.667$**	1×
39	Write the formula for y', y'' from Newton-Gregory forward	$dy/dx = (1/h)[\Delta y_0 + (2p-1)/2 \cdot \Delta^2 y_0 + (3p^2-6p+2)/6 \cdot \Delta^3 y_0 + \ldots]$ $d^2y/dx^2 = (1/h^2)[\Delta^2 y_0 + (p-1)\Delta^3 y_0 + (12p^2-36p+22)/24 \cdot \Delta^4 y_0 + \ldots]$ At $p=0$: **$y' = (1/h)[\Delta y_0 - \Delta^2 y_0/2 + \Delta^3 y_0/3 - \ldots]$**	2×
40	Determine $\Delta^2 y_2$, $\Delta^3 y_1$ from $y = 0.0567, 1.2345, 1.5678, 1.0023, 1.786$	Δ: 1.1778, 0.3333, −0.5655, 0.7837 Δ^2: −0.8445, −0.8988, 1.3492 Δ^3: −0.0543, 2.2480 **$\Delta^2 y_2 = -0.8988$, $\Delta^3 y_1 = -0.0543$**	1×

UNIT V — Q9(b) or Q10(b)

#	QUESTION	WORKED ANSWER	SEEN
41	**Gauss-Seidel**: $2x+17y+4z=35$; $3x+y+10z=24$; $28x+4y-z=32$, from (0,0,0)	**Reorder for dominance:** $28x+4y-z=32$; $2x+17y+4z=35$; $3x+y+10z=24$ $x=(32-4y+z)/28$, $y=(35-2x-4z)/17$, $z=(24-3x-y)/10$ It1: 1.142857, 1.924370, 1.864706 **It2: $x=0.934544$, $y=1.510123$, $z=1.968625$** (→ 1.001, 1.482, 1.952)	1×
42	**Gauss-Seidel**: $5x+2y+z=12$; $x+4y+2z=15$; $x+2y+5z=20$, from (1,0,0)	Already diagonally dominant. $x=(12-2y-z)/5$, $y=(15-x-2z)/4$, $z=(20-x-2y)/5$ It1: 2.400000, 3.150000, 2.260000 **It2: $x=0.688000$, $y=2.448000$, $z=2.883200$** (exact solution 1, 2, 3)	1×
43	**Rayleigh's power method**: $A = [[2,-2,1],[1,3,-1],[-1,2,2]]$, $X_0=[1,0,0]^T$, 2 iterations	$AX_0 = [2,1,-1]^T = 2[1, 0.5, -0.5]^T$ → $\lambda_1 = 2$ $AX_1 = [0.5, 3, -1]^T = 3[0.1667, 1, -0.3333]^T$ → **$\lambda_2 = 3$, $X = [0.167, 1, -0.333]^T$** *Careful: this matrix has **complex** eigenvalues (2.397±2.163i), so the method never actually converges. The question only asks for two iterations — write them and stop.*	1×
44	Find x so that $A = [[4-x, 2\sqrt{5}, 0],[2\sqrt{5}, 4-x, \sqrt{5}],[0, \sqrt{5}, 4-x]]$ has **rank 2**	Rank < 3 requires $\|A\| = 0$. $\|A\| = -(x-9)(x-4)(x+1)$ **$x = 9, 4$ or -1** *(check a 2×2 minor is non-zero at each, so the rank is exactly 2)*	1×
45	Prove: if λ is an eigenvalue of A then $1/\lambda$ is an eigenvalue of A^{-1}	$AX = \lambda X$. Pre-multiply by A^{-1}: $A^{-1}AX = \lambda A^{-1}X$ → $X = \lambda A^{-1}X$ Divide by λ (non-zero since A is invertible): **$A^{-1}X = (1/\lambda)X$** ■	1×

Three that bite.
1. Gauss-Seidel — reorder first. Both systems above need it (or check they are already dominant). Charge straight in and the numbers wander.
2. Missing terms — count your knowns. 3 knowns → quadratic → set 3rd differences to zero. 5 knowns → quartic → 5th differences. Setting one higher difference to zero gives one equation and you cannot solve for two unknowns.
3. Cooling — do not compute k. Square (or cube) the ratio instead: if $e^{-10k} = 17/22$ then $e^{-20k} = (17/22)^2$. Faster and no rounding error.

Why this bank is worth more than it looks

20 marks, and each question is ~6 minutes. Combined with the 2-mark bank that is **30 marks in the first 50 minutes of the exam**, before you touch a single 7-marker.
Unit III is free: `1/(D+a)` has been the `Q5(b)` part in **3 of 3** exact-match papers. Memorise the four-line proof and that unit's 4-marker is done.
Show the working. These are method-marked — a Gauss-Seidel table with a slip in the last row still scores 3 of 4; a bare answer scores 1.

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45 questions: every 4-mark part asked across the 6 papers, plus the standard forms the portion supports. Every numerical result was computed and verified — including the quartic for #32 and the eigenvalues for #43. Counts are appearances across those papers; LIKELY means not yet asked but squarely on the syllabus.